
MSP: Macro-level Soccer Predictor

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Abstract

Soccer is the most popular sport worldwide and constitutes the fastest growing market in sports gambling. As a result, a substantial body of research has emerged aiming to forecast football match outcomes using various statistical and machine learning methods. In this paper, we explore an alternative approach by applying probabilistic graphical models (PGMs) to predict the outcome of individual soccer matches. Specifically, we investigate both discriminative and generative modeling strategies, with a focus on latent variable models that capture unobserved team characteristics. Our model aims to provide interpretable and flexible predictions by explicitly modeling the relationships between observable features and latent factors such as team strength.

1. Introduction

Accurately predicting the outcome of a soccer match is a notoriously difficult task. The low-scoring nature of the sport, combined with unpredictable contextual and psychological factors, makes reliable forecasting challenging. Despite this, the global popularity of soccer and the rapid growth of the sports betting industry have fueled a surge in research interest. A wide range of statistical and machine learning models have been applied to match outcome prediction — from classical models such as logistic regression and random forests (Wong et al., 2025; Arntzen & Hvattum, 2020; Robberechts & Davis, 2019), to Bayesian models (Constantinou et al., 2012), and more recently, deep neural networks like RNNs and LSTMs (Jain et al., 2021; Rahman, 2020; Danisik et al., 2018).

While highly accurate models are essential for bookmak-

ers and bettors, other stakeholders — such as coaches, analysts, and team management — have a pressing need for interpretability. Understanding which match features contribute most to success can inform tactical decisions, player selection, training priorities, and long-term strategic planning. However, many recent deep learning-based approaches operate as “black boxes”, limiting their utility in practice despite strong performance.

In this project, we focus on forecasting individual match outcomes (home win, draw, away win) with an emphasis on interpretable modeling. Our approach builds on the work of Constantinou et al. (2012), extending their Bayesian framework into the domain of latent variable graphical models. These models allow us to explicitly represent complex dependencies between observed match features (e.g., recent performance, home advantage) and latent factors such as team strength.

At the core of our model are two key latent variables: the strength of the home team and the strength of the away team. These are inferred from a range of sub-features that reflect multidimensional team performance. The predicted outcome is then derived by comparing these latent strength variables in a probabilistic framework.

By structuring the model in this way, we aim not only to make accurate predictions but also to uncover how different features interact to influence outcomes. The modular nature of probabilistic graphical models (PGMs) further allows us to flexibly incorporate new data sources and extend the model as needed.

Ultimately, this work contributes to the development of predictive systems that are both accurate and explainable. Beyond applications in betting, our model has potential uses in real-time match analysis and as a decision-support tool for coaching staff and performance analysts.

2. Related Work

To motivate our use of PGMs for soccer match prediction, we surveyed several influential papers in this field.

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2.1. Statistical and Probabilistic Approaches

Early football prediction models established the foundation for modern approaches using statistical methods. Maher (1982) introduced Poisson distribution-based models for predicting match scores, while Dixon & Coles (1997) refined this approach by incorporating time-dependent factors and demonstrated profitability against betting markets. These foundational works established the importance of probabilistic modeling in football prediction, which directly relates to our Bayesian network approach using latent variables to capture team strength.

2.2. Bayesian Network Frameworks

Constantinou et al. (2012) developed "pi-football," a comprehensive Bayesian network model that demonstrated superior performance through hierarchical modeling integrating team strength metrics, historical performance, and subjective factors. Their approach achieved a Ranked Probability Score (RPS) of 0.1319 and generated 9.48% ROI against bookmaker odds. This work is particularly relevant to our project as it validates the effectiveness of Bayesian networks for football prediction and establishes the importance of team strength as a latent variable, which aligns with our graphical model design incorporating Team Strength (Home) and Team Strength (Away) nodes.

Robberechts et al. (2021) introduced a time-dependent Bayesian framework for in-game win probability estimation, achieving an average RPS of 0.134. Their approach of modeling future goal scoring as independent Poisson processes with scoring intensities estimated from game state features provides methodological insights for our macro-level prediction framework, particularly in handling temporal dynamics through historical performance data.

2.3. Event-Based Machine Learning

Recent advances in event data utilization have enhanced prediction accuracy. Hassard & Kerr (2024) implemented XGBoost and random forest classifiers using StatsBomb event data, achieving 65.8% accuracy with SVM. Wong et al. (2025) proposed a two-stage stacking ensemble combining CNN and LightGBM, incorporating meteorological variables and achieving 67.2% accuracy. While these approaches focus on micro-level event data, our macro-level approach using Transfermarkt data provides a complementary perspective by emphasizing market valuations, team composition metrics, and aggregate performance indicators.

Collectively, these studies substantiate our selection of a PGM framework due to its inherent interpretability, flexibility in combining subjective and objective information, and effectiveness in capturing complex statistical dependencies.

dencies.

3. Proposed Methods

Approaches to soccer outcome prediction are commonly divided into three categories (Constantinou, 2019): (1) statistical models, (2) machine learning and probabilistic graphical models, and (3) rating systems. In our work, we propose two distinct probabilistic modeling approaches for predicting football match outcomes based solely on pre-match features.

First, we introduce a *Latent Bayesian Network (Latent BN)* that incorporates unobserved "team strength" variables to simplify relationships between observable features and outcomes. This approach combines elements from both category (2) and category (3), as it uses a graphical model framework to infer latent team ratings from observable attributes.

Second, we implement a *Bayesian multinomial logistic regression model (Flat Logit)* that directly maps continuous pre-match features to outcome probabilities. This model falls under category (1) and serves as a more sophisticated baseline, offering a strong probabilistic alternative with improved granularity and inference efficiency. Beyond its role as a baseline, it also allows us to assess the potential benefits of using continuous input features - unlike the Latent BN model, which relies on discretized representations.

3.1. Proposed Model

We model the outcome of a soccer match using a probabilistic graphical model (PGM) that captures statistical dependencies between match- and team-level variables (Figure 1).

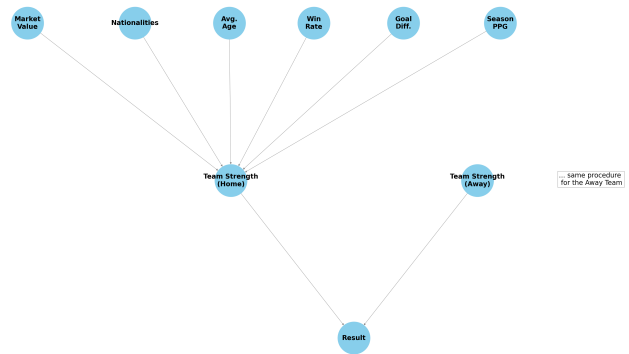


Figure 1. Structure of the proposed probabilistic graphical model

At the core of the model are two latent variables: *Team Strength (Home)* and *Team Strength (Away)*. These variables are independently inferred for each team based on observable features that capture historical performance and

team composition. A detailed description of the selected input features can be found in Section 4.1.1. These variables are aggregated to form a composite latent representation of team strength. The same process is applied separately for the home and away teams. The match result (home win (H), draw (D), or away win (A)) is then modeled as dependent on both team strength variables. This allows the model to capture how relative team strength influences the outcome.

Since both the features and the latent variables are continuous, we apply discretization to enable the use of a Bayesian network. Specifically, we discretize the features using a binning approach, where the number of bins for the features b and the number of clusters for the latent variables c serve as hyperparameters. These hyperparameters have a significant impact on the model’s performance. Through hyperparameter tuning, we observed that increasing the binning granularity for both the features and the latent variables generally improves the quality of the posterior distributions over the latent variables. In particular, it enhances the conditional probabilities $P(Y | X)$, where $Y \in \mathbb{R}^c$ denotes the latent variable representing home or away team strength, and $X \in \mathbb{R}^{d \times b}$ represents the discretized feature set where d is the number of features. However, increasing the binning resolution for features and latent variables comes at the cost of significantly longer training times.

4. Experiments

4.1. Dataset

For our project, we selected Transfermarkt (Transfermarkt, 2025) as our primary dataset. The dataset is recent, well-maintained, and offers structured macro-level data covering football players, clubs, matches, and transfers over time. It is relational in structure, updated weekly, and includes multiple leagues and seasons, with over 1.7 million unique player appearances. Additional information includes club-level details, match outcomes, player metadata, and historical market valuations, which can be used to track team value changes over time.

We performed preprocessing steps including table joins, filtering by competition and season, and calculating and normalizing team-level features such as average market value and player statistics.

The label distribution of match outcomes in our final dataset is shown in Table 1. As anticipated, home wins are the most common, whereas draws are the least frequent result.

4.1.1. FEATURE SELECTION

Figure 2 reports pair-wise Pearson correlations. Whenever two columns exceed the $|\rho| > 0.75$ threshold we elimi-

Match Outcome	Count	Percentage
Home Win (0)	26,891	44.5%
Draw (1)	14,285	23.6%
Away Win (2)	18,429	30.5%
Total	59,605	100%

Table 1. Label distribution in the final dataset, both in absolute counts and percentages.

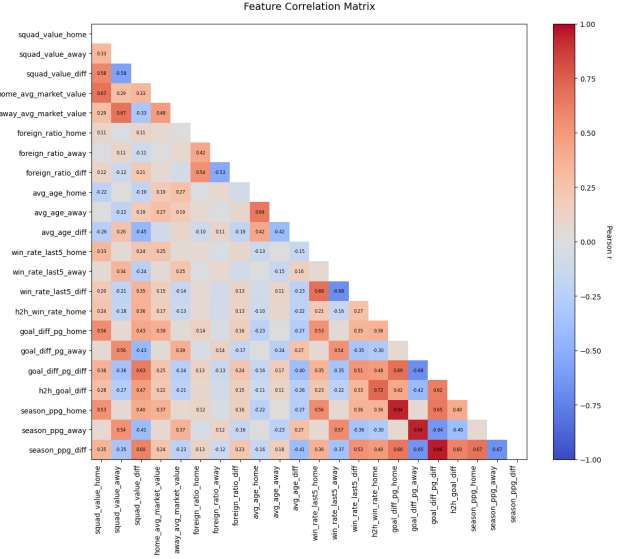


Figure 2. Pearson correlation matrix for the 22 engineered features.

nate the one that is either (i) less interpretable or (ii) largely duplicated by another retained metric. Three columns are removed. We drop **goal_diff_pg** * ($\rho = 0.94$ with **season_ppg** *), while **h2h_goal_diff** is kept. We also drop **squad_value** *, ($\rho = 0.67$ with ***_avg_market_value**), as they are essentially calculated from same data with different equations, but ***_avg_market_value** represents the valuation of players on that specific match day.

The “_home”, “_away”, and “_diff” variants that remain are kept intentionally: the directional signal (e.g. strength differential) is required. After pruning, 16 features are selected for learning pipeline; their grouping is summarized in Table 2.

4.2. Baseline Models

To evaluate the added value of our graphical model, we compare it against several standard baseline models from classical machine learning, as well as our more sophisticated probabilistic baseline. All these models predict match outcomes directly from the observable input features, without incorporating any latent variables. Conceptually, this corresponds to a simplified version of our graphical model (Figure 1), where the latent nodes *Team Strength*

Feature	Corresponding columns	Description
Market value	*_avg_market_value	Average market value of starting 11
Nationalities	foreign_ratio_*	Ratio of foreign players in each team
Avg. age	avg_age_*	Average squad age on game day
Win rate	win_rate_last5_*, h2h_win_rate	Recent (5 matches) and head-to-head win percentages
Goal diff.	h2h_goal_diff	Average goal difference in head-to-head matches
Season PPG	season_ppg_*	Points per game in the current season

Table 2. Mapping of paper-level features to concrete columns (* ∈ {home, away, diff}).

(*Home*) and *Team Strength (Away)* are omitted.

4.2.1. SIMPLE BASELINE MODELS

As standard baseline models, we employ logistic regression, random forest, and XGBoost. Despite their simplicity, these models already achieve performance comparable to results reported in related work, where similar approaches reach accuracies between 59% and 65% (Wong et al., 2025). Detailed evaluation metrics - including per-class precision, recall, and overall accuracy - are presented in Table 3.

Model	Metric	Class 0	Class 1	Class 2	Accuracy
LogReg	Precision	0.5411	1.0000	0.5104	0.5317
	Recall	0.8276	0.0007	0.5114	
RF	Precision	0.5473	0.3028	0.5011	0.5213
	Recall	0.7845	0.0613	0.4940	
XGB	Precision	0.5564	0.3056	0.5122	0.5283
	Recall	0.7745	0.0721	0.5228	

Table 3. Per-class precision and recall, along with overall accuracy, for the baseline models.

4.2.2. FLAT BAYESIAN LOGISTIC REGRESSION

As mentioned, as a more sophisticated baseline, we employ the flat Bayesian logistic regression model. In contrast to our graphical model, this approach retains a simple structure but operates directly on continuous input features instead of discretized ones. The goal is to evaluate whether leveraging continuous representations can improve predictive performance while preserving a simple model structure.

Model. Given a design matrix $\mathbf{X} \in \mathbb{R}^{N \times K}$ of standardized, continuous match features, we model the categori-

cal outcome $y_n \in \{0, 1, 2\}$ (Home, Draw, Away) with a reference-class multinomial logistic regression:

$$p(y_n = j \mid \mathbf{x}_n, \mathbf{B}, \mathbf{b}) = \text{softmax}([0, \mathbf{x}_n^\top \mathbf{B}] + \mathbf{b})_j, \quad (1)$$

$$j \in \{0, 1, 2\}.$$

where the first (*Home-win*) class acts as baseline. The coefficient matrix $\mathbf{B} \in \mathbb{R}^{K \times 2}$ and intercept vector $\mathbf{b} \in \mathbb{R}^2$ receive weakly informative priors $\mathbf{B}_{ik} \sim \mathcal{N}(0, 0.5^2)$ and $b_k \sim \mathcal{N}(0, 0.5^2)$.

To mitigate class imbalance we weight each data point’s contribution to the log-likelihood by the inverse class frequency, $w_n = \max_c N_c / N_{y_n}$. But this has similar effect to soft EM, a tradeoff between accuracy and draw prediction.

For inference we employ the No-U-Turn Sampler (NUTS) in PYMC, running *two* chains with 1 000 warm-up and 1 000 posterior iterations each. Convergence is satisfactory: every $\hat{R} < 1.01$ and the BFMI lies in the interval 0.91–1.34; no divergences were recorded. The full run completes in ≈ 20 min on a MacBook M3 Pro.

We keep all posterior draws of $(\mathbf{B}, \mathbf{b})$ and the corresponding predictive probabilities $\mathbf{p}_n = \text{softmax}([0, \mathbf{x}_n^\top \mathbf{B}] + \mathbf{b})$; test-set accuracy is evaluated on the posterior-mean predictions.

Results. On the 20% hold-out split the Bayesian flat-logit attains an accuracy of 0.506 on *balanced dataset* and a accuracy of 0.537 on *unbalanced dataset*, sacrificing class 1 (Draw) prediction. Table 4 lists the full per-class scores: the model is most confident on the *Home-win* outcome (precision/recall 0.642/0.571) and least on *Draw* (0.295/0.305). The resulting accuracy of 0.506 exceed both unbalanced soft-EM (0.478) and balanced soft-EM (0.407).

Model	Metric	Class 0	Class 1	Class 2
Flat Logit	Precision	0.642	0.295	0.500
	Recall	0.571	0.305	0.567
Accuracy		0.506		

Table 4. Per-class precision and recall, along with overall accuracy, for the Flat Logit model on the balanced dataset.

Comparison with Baseline and Existing Methods.

Comparing our Bayesian flat-logit with three standard baselines, namely maximum-likelihood logistic regression (LOGREG), random forest (RF), and gradient-boosted trees (XGB) (Table 3), all baselines strongly favour the dominant *Home-win* class, showing high recall there while almost ignoring the rare *Draw* outcome. The Bayesian model trades a small drop in class-0 recall for markedly higher precision and recall on both minority classes,

achieving a far more balanced error profile while almost matching the best baseline in overall accuracy.

Additionally, traditional frequentist logistic regression, although equally interpretable, cannot quantify posterior uncertainty, and tree ensembles yield only point-estimate probabilities. Purely discrete Bayesian networks remain attractive for small feature sets but require carefully crafted structures to scale. Our Bayesian flat-logit retains interpretability, while realistically trading extra computation for predictive performance.

4.3. Latent Variable Model

One of our core modeling approaches is a latent variable model, for which we employ the Soft-EM algorithm to perform parameter estimation. We focus primarily on the Soft-EM variant rather than its hard counterpart, as initial experiments showed that it consistently achieved slightly better performance in terms of predictive accuracy.

4.3.1. INITIALIZATION

Simple random initialization of the model parameters often led to zero precision and recall for one or more classes, suggesting that the model had converged to poor local optima and overfit to specific outcome classes in the dataset. To address this, we chose KMeans-based initialization, which provides a more structured and stable starting point. This approach consistently outperformed both single and multi random initialization across all key evaluation metrics - accuracy, precision, and recall - for all three outcome classes. Moreover, compared to multi random initialization, it achieved better performance with lower computational cost, making it a more efficient and effective choice.

In the Soft-EM algorithm, we fixed the number of iterations to 5, as we observed that the model typically converges within the first few steps. To determine appropriate bin sizes for feature discretization and latent variable states, we performed a grid search to balance granularity and computational efficiency. We ultimately selected a feature binning size of $b = 5$ and a latent variable binning size of $c = 5$. Note that the number of clusters k used for KMeans initialization is always set to match the number of latent variable bins, i.e., $k = c$, to ensure consistency between initialization and model structure.

4.3.2. RESULTS

Using the dataset, we obtain the evaluation metrics shown in Table 5.

As seen in the results, both precision and recall for class 1 (draw) are 0. This indicates that the model completely fails to predict draws. A likely reason for this is the heavy class imbalance in the original dataset, where draws occur much

Model	Metric	Class 0	Class 1	Class 2
Soft-EM	Precision	0.4832	0.0000	0.4469
	Recall	0.8855	0.0000	0.2408
Accuracy			0.4772	

Table 5. Per-class precision and recall, along with overall accuracy, for the Soft-EM model.

less frequently than home or away wins. Consequently, the model learns to favor the majority classes and largely ignores the minority class.

To address this issue, we apply class balancing by adjusting the dataset such that all outcome labels are equally represented. This results in 14,285 samples per class and leads to the following performance:

Model	Metric	Class 0	Class 1	Class 2
Soft-EM	Precision	0.4317	0.3574	0.4229
	Recall	0.4291	0.3098	0.4817
Accuracy			0.4072	

Table 6. Per-class precision and recall, along with overall accuracy, for the Soft-EM model on the balanced dataset.

Thus, we observe a significant trade-off between model accuracy and class-wise performance when balancing the dataset. As shown in Table 5 (unbalanced), the Soft-EM model achieves a higher overall accuracy but fails to predict draws (class 1), with precision and recall values at 0. This reflects the strong class imbalance in football outcomes, where draws are less frequent than home or away wins.

In contrast, balancing the dataset (see Table 6) leads to lower overall accuracy, but significantly improves performance on the minority class. The model learns to recognize draws more reliably, with non-zero precision and recall values across all classes.

Depending on the use case, one might justify ignoring draws if the goal is purely win/loss prediction. However, for general-purpose or analytical applications where all outcomes are of interest, balancing the data provides a more robust and fair model.

4.3.3. INTERPRETING LATENT TEAM STRENGTH

The latent variables *HomeStrength* and *AwayStrength* play a central role in our graphical model. Although these variables are not directly observed, they are inferred from observable team features using an unsupervised clustering step (KMeans), followed by iterative soft-EM refinement.

To facilitate a meaningful interpretation of the latent variables *HomeStrength* and *AwayStrength*, we reassigned the

initial cluster labels based on a composite score derived from all selected features. All features were normalized to the range $[0, 1]$ using MinMax scaling to ensure comparability across dimensions and to prevent any single feature from disproportionately influencing the cluster ranking. The composite score was then calculated as the sum of all normalized feature values - implicitly treating all variables as positively associated with team strength.

This approach enabled a consistent relabeling of clusters from weakest (Cluster 0) to strongest (Cluster 4) based on aggregated, normalized feature scores. The validity of this relabeling is supported by the average feature profiles shown in Tables 7 to 9: By considering all features, Cluster 4 demonstrates the highest values in total — reflecting strong and competitive teams. In contrast, Cluster 0 exhibits comparatively low values in most dimensions, suggesting weak squads with limited recent success.

HomeStrength	Market Value	Nationalities	Avg. Age
0	0.0104	0.7872	1.83
1	0.0053	2.1480	1.93
2	0.0313	1.5568	1.73
3	0.0570	2.4608	1.92
4	0.1470	2.1370	1.90

Table 7. Squad-related features per HomeStrength cluster.

HomeStrength	Win Rate	Goal Diff.	Season PPG
0	1.56	2.06	1.78
1	0.51	1.99	1.34
2	1.74	2.07	1.86
3	1.83	2.10	2.08
4	3.99	2.40	2.64

Table 8. Performance-related features per HomeStrength cluster.

HomeStrength	Rest Days
0	0.69
1	0.80
2	3.92
3	0.58
4	0.52

Table 9. Rest time per HomeStrength cluster.

Therefore, Cluster 0 represents the weakest teams (low quality and output), Clusters 1–3 cover varying intermediate profiles, and Cluster 4 captures the strongest, most disciplined teams.

To further validate this interpretation, we examine the conditional probability table (CPT) of the match outcome given both *HomeStrength* and *AwayStrength*. Instead of listing selected rows, we visualize the full matrix of $P(\text{Home Win} \mid \text{HomeStrength}, \text{AwayStrength})$ as a heatmap in Figure 3.

The pattern is largely consistent with intuition: high probabilities occur when strong home teams face weaker away teams (e.g., $P(\text{Home Win} \mid \text{HomeStrength} = 4, \text{AwayStrength} = 1) = 0.89$), while the lowest values appear when weak home teams face strong opponents (e.g., $P(\text{Home Win} \mid \text{HomeStrength} = 0, \text{AwayStrength} = 4) = 0.20$).

However, small inconsistencies remain. For instance, $P(\text{Home Win} \mid \text{HomeStrength} = 4, \text{AwayStrength} = 1) = 0.89$ is slightly higher than $P(\text{Home Win} \mid \text{HomeStrength} = 4, \text{AwayStrength} = 0) = 0.83$, despite the latter involving a weaker away team — which is counterintuitive.

We assume that such artifacts result from coarse discretization of both feature values and latent variable states. Increasing the number of bins for continuous features and enhancing the granularity of latent variable representations may help mitigate these effects.

Still, this confirms that the latent strength variables learned by the model generally have a meaningful impact on the predicted outcome.

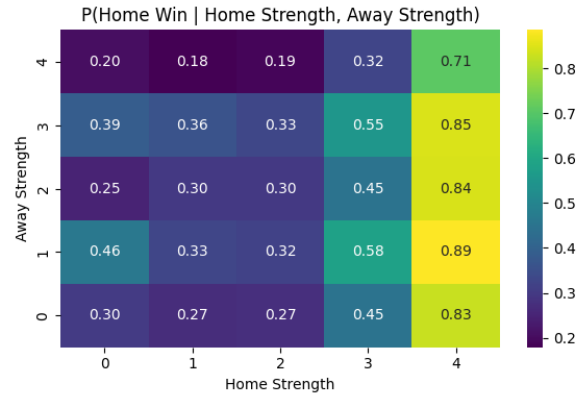


Figure 3. Heatmap based on the final Bayesian model.

Consequently, unlike black-box models, our graphical approach introduces structure and interpretability through the latent variables *HomeStrength* and *AwayStrength*, which mediate between observed features and predicted match outcomes.

These latent variables act as semantic abstractions of team quality. Using unsupervised clustering and soft-EM refinement, the model groups teams into meaningful strength categories based on the selected features. As shown in Tables 7 to 9, each cluster exhibits interpretable patterns — for example, Cluster 4 reflects high-quality, disciplined teams with strong performance metrics, while Cluster 0 groups weaker teams with poor output.

This design enables two levels of interpretation:

- **Feature-to-latent mapping:** For any given team sample, we can identify which strength cluster it was assigned to (e.g., *HomeStrength* = 3) and interpret this label via the cluster’s average features. This supports statements like “the model considers the home team strong due to its recent win rate and high squad value.”.
- **Latent-to-prediction reasoning:** The prediction itself is based on the learned conditional probability table, $P(\text{MatchOutcome} \mid \text{HomeStrength}, \text{AwayStrength})$, which explicitly encodes how combinations of team strengths affect outcome probabilities. For instance, strong home teams against weak away teams yield high home win probabilities.

Together, this structure enables transparent and diagnostic analysis: users can trace a prediction from input features to latent abstractions and finally to output probabilities. This supports not only improved understanding, but also informed decision-making in practical applications such as match analysis or betting strategies. So, while the model may not outperform our black-box baselines in terms of raw accuracy, its strength lies in the added value of interpretability - making it particularly suitable when understanding the underlying decision rationale is as important as the prediction itself.

5. Conclusion

This study on macroscopic soccer match prediction models reveals critical trade-offs between accuracy, efficiency and interpretability. Utilizing all features as continuous variables maximizes prediction accuracy by capturing subtle match patterns and team capabilities. However, this approach significantly increases training time, creating computational constraints for real-world implementation. Future research should focus on feature selection and dimensionality reduction techniques that maintain predictive power while reducing complexity. Distributed computing and algorithm optimization warrant further investigation.

The comparable performance between EM algorithm approximation and exact Bayesian computation indicates that data uncertainty imposes fundamental limits on prediction performance regardless of computational sophistication. In such cases, interpretable models like our latent-variable PGM offer a practical advantage: they allow users to trace predictions through structured abstractions such as team strength, providing insight even when perfect accuracy is unattainable. This makes efficient approximation techniques such as EM particularly appealing, balancing tractability with meaningful explanation.

Overall, our work highlights that probabilistic graphical

models (PGMs) are well-suited to macro-level prediction tasks in soccer: they flexibly combine heterogeneous information, support modular extensions, and expose interpretable latent structures. Rather than aiming solely for maximum predictive performance, future modeling efforts should seek a principled trade-off between performance, efficiency, and the ability to explain predictions in a way that supports real-world decision-making.

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